

SignTalk AI: Complete System & Developer Guide

Bidirectional AI Communication Bridge for Hearing & Speech Impaired

Department: Information Science & Engineering, RNS Institute of Technology
Academic Year: 2025–2026 | **Batch:** #24 | **Guide:** Umesh M
Team: Arjun B Shetty, Khushi Ramesh, Mahin S Kunder, Sathya Shriya G N
Status: Production prototype live. Ready for final validation, tuning, and IEEE paper integration.

1. Executive Summary & Problem Statement

Over 466 million people globally suffer from disabling hearing loss. Conventional communication tools either rely on specialized hardware or awkward back-and-forth typing. **SignTalk AI** eliminates this friction through a dual-channel multimodal AI platform:

- **Sign Language to Spoken Voice:** Translates continuous Indian Sign Language (ISL) from a standard webcam into natural grammatical speech with facial emotion awareness.
- **Speech to Sign Subtitles:** Transcribes the hearing user's voice into real-time visual subtitles with Whisper AI.
- **Multilingual & Context Aware:** Translates dialogues between English, Hindi, and Kannada, backed by Gemini 2.0 Flash for sentence reconstruction.
- **Zero-Network Resilience (Airplane Mode):** Runs entirely on-device when connectivity drops, using quantized TFLite models, local Flan-T5, and offline phrasebooks.

2. End-to-End Pipeline Architecture

The system coordinates 6 modular machine learning stages across asynchronous WebSocket pipelines:

Stage	Model / Library	Function & Specification
1. Pose Extraction	MoveNet Thunder (TF Hub)	Extracts 17 keypoints (body joints, shoulders, wrists). Coordinates normalized relative to shoulder midpoint and bounding-box diagonal.
2. Sequence Classifier	BiLSTM (Keras / TFLite)	Evaluates 30-frame rolling sequence (shape: 30x17x2). Trained on INCLUDE dataset (40 curated classes, 89.4% val accuracy). Debounced across 3 agreeing frames.
3. Emotion Detection	DeepFace (7 classes)	Runs on every 10th frame to minimize CPU overhead. Classifies emotion: happy, sad, angry, fear, surprise, neutral, disgust.
4. NLP Sentence Fixer	Gemini 2.0 Flash / Flan-T5	Fuses recognized gesture keywords + dominant emotion + conversation history into a natural grammatical sentence. Falls back to local Flan-T5.

5. Multilingual Translation	Google Translate REST / Cache	Translates sentences to English, Hindi, or Kannada. Uses local LRU cache and offline phrasebook fallback.
6. Speech & Synthesis	Whisper Small & gTTS / Coqui	Whisper Small transcribes hearing voice; gTTS/Coqui synthesizes translated text into spoken audio.

3. Locked Contracts (NEVER MODIFY WITHOUT FULL-TEAM REVIEW)

These contracts link the frontend, backend, ML models, and mobile app together. Changing function arguments or array dimensions will break other layers:

- **1. Exactly 17 Keypoints (MoveNet Thunder):** The IEEE paper mentions 33 keypoints (MediaPipe), but our entire production ML pipeline is locked to **17 keypoints**. Do NOT switch to MediaPipe without retraining everything.
- **2. Classifier Signature:** `classify_sequence(sequence: np.ndarray)` takes exactly shape (30, 17, 2) and returns `{'label': str, 'confidence': float}`.
- **3. WebSocket Protocol (/ws/gesture):** Client streams JPEG frames at 20–30 FPS. Backend stabilizes output: requires **3 consecutive agreeing predictions** and a **1.5s cooldown** before re-emitting the same sign.
- **4. Unmirrored Webcam Rule:** In `frontend/src/components/WebcamView.jsx`, `mirrored={true}` must remain disabled. Mirroring flips wrists horizontally, which ruins recognition against the unmirrored training dataset.
- **5. Frame Skip Setting (?frame_skip=2):** Live recognition WebSocket must connect with `?frame_skip=2`. Because real signs take ~2.5–3s, at 20 FPS a 30-frame window with `skip=2` spans ~3.0s, matching the training clip duration.
- **6. Shared Offline Mode:** All services check `X-Offline-Mode: true` header or `?offline=true` query param through `api.core.offline_mode.get_offline_mode`.

4. Repository Structure & File Guide

Directory / File	Role & Responsibility
backend/api/	Production FastAPI backend: routes for pose, emotion, AI, speech, translation, analytics.
backend/keypoint_utils.py	MoveNet extraction, spatial normalization, temporal smoothing buffer.
backend/classify.py	Central gesture classification logic loading the active BiLSTM model.
backend/nlp_correction.py	Gemini 2.0 Flash prompt runner with local Flan-T5-small fallback.
backend/emotion.py	DeepFace facial emotion recognition and 2-second majority voting filter.
backend/translation.py	Translation engine: in-memory cache -> Firestore -> Google API -> phrasebook.
backend/train_bilstm.py	BiLSTM training script with stratified train/val split and TFLite exporter.
backend/runs/	Trained model weights (active: <code>exp_top40_stratified</code> , 89.4% accuracy).
frontend/src/	React web dashboard: WebcamView, SubtitleBar, LanguageSelector, AnalyticsPanel.
mobile/	Flutter app (cross-platform client sharing the same WebSocket/REST contracts).

5. Step-by-Step Developer Quickstart

Terminal 1 — Backend (FastAPI + WebSocket on Port 8000):

```
cd backend
.\venv\Scripts\activate
uvicorn api.socket_manager:socket_app --host 127.0.0.1 --port 8000 --reload
```

Terminal 2 — Frontend (Vite React on Port 3000):

```
cd frontend
npm run dev
```

- Frontend URL: <http://localhost:3000>
- Swagger Docs: <http://127.0.0.1:8000/docs>

6. Immediate Gotchas & Action Items for the Week

To finish the project within the next few days, focus exclusively on these key fixes:

Priority Item	Severity	Action Required
1. TFHub MoveNet Cache Corruption	HIGH	If Uvicorn crashes at startup saying 'saved_model.pb not found', clear the directory %TEMP%\tfhub_modules. The script will automatically re-download a fresh copy.
2. Cloud Translation API Key & Billing	HIGH	Google Cloud Translation API requires billing. If inactive, the system gracefully falls back to phrasebook.json. Either provide a billing-enabled key or expand phrasebook.json for demo phrases.
3. Firebase Service Account & Mock Mode	MEDIUM	Ensure backend/secrets/firebase-service-account.json is present. If testing without GCP credentials, verify that the firebase client's mock fallback is enabled so Auth and Firestore calls don't block login.
4. Fine-Tuning Sentence Correction Prompts	MEDIUM	Check backend/nlp_correction.py prompt templates. Ensure Gemini returns concise sentences that sound natural and preserve the emotional tone.
5. Latency & Demo Preparation	HIGH	Verify that end-to-end latency remains under 300ms. Prepare 5 standard demo gestures from the 40-word vocabulary (e.g. HELLO, THANK YOU, PLEASE, HELP, GOOD) with clear lighting.

7. Datasets & Model Training Reference

- **Dataset:** INCLUDE (Indian Sign Language dataset). All 15 categories preprocessed into 4,257 sequence files (backend/data/sequences/).
- **Curated Vocabulary:** 40 top-represented classes with stratified train/validation split achieving **89.4% accuracy**.
- **Training Command:** `python train_bilstm.py --data_dir data/sequences --labels_csv data/labels_top40.csv --split_mode stratified --output_dir runs/exp_top40_stratified`
- **TFLite Conversion:** `python convert_to_tflite.py --saved_model runs/exp_top40_stratified/saved_model --output models/bilstm.tflite`

Pro-Tip for Handover: Have your friend start with Section 5 (Quickstart), run both terminals, and test the Swagger UI at <http://127.0.0.1:8000/docs>. The system is modular: frontend UI, gesture pipeline, and NLP correction can each be debugged and modified independently.